Predicting Linear Relational Decodability Without the Relation

Anonymous ACL submission

Abstract

Linear relational embeddings (LREs) approximate a language model's subject-to-object mapping with an affine function, and their faithfulness varies enormously across relations—near zero for PERSON_FATHER, at ceiling for COUNTRY_CAPITAL_CITY. We ask what determines that variation, and show it is substantially predictable from a readout that never sees the relation. We estimate a Jacobian lens from subject-to-subject differences under a single neutral prompt, at one fixed layer for every relation, using no relation template, no objects and no (subject, object) pairs, and measure how highly it ranks the model's own answer. Across 28 relations of the LRE dataset on Olmo-3-7B this relation-agnostic score correlates with LRE faithfulness at $\rho = 0.66$ ($p = 1.4 \times 10^{-4}$), rising to 0.75 once object cardinality is partialled out. Neither cardinality nor the model's prior over the answer token accounts for the correlation, and the estimate is stable under leave-one-domain-out (0.57–0.70). Linear decodability thus appears to be largely a property of what a subject's representation already carries, rather than of the relation being decoded.

1 Introduction

Hernandez et al. (2024) showed that a language model's mapping from a subject to the object of a relation is often well approximated by an affine function of the subject's hidden state—a *linear relational embedding* (LRE). They also showed that this works far better for some relations than others. On Olmo-3-7B we reproduce the same spread: an LRE fitted on COUNTRY_CAPITAL_CITY agrees with the model's own prediction 75% of the time, while one fitted on PERSON_FATHER agrees 9% of the time.

Why? The natural reading is that some relations are "more linear" than others—that linearity is a property of the relation. We report evidence that it is better understood as a property of the *subject's representation*. Concretely, we show that one can predict how faithful an LRE will be *before fitting it*, and without using the relation at all.

Our diagnostic is a Jacobian lens estimated from differences between subject representations under a single neutral prompt, shared by every relation. It sees no relation template, no objects, and no (subject, object) pairs; the layer is fixed in advance rather than tuned per relation. We then ask how highly this lens ranks the model's own answer to the relation. That score correlates with LRE faithfulness across 28 relations at $\rho = 0.66$.

We make three contributions. (i) We introduce a relation-agnostic diagnostic of linear decodability and show it predicts LRE faithfulness. (ii) We show the prediction survives controls for object cardinality and answer-token frequency, and for holding out entire subject domains. (iii) We document a first-token scoring artifact that silently inflates faithfulness to a perfect score on relations with numeric objects, and give the fix.

2 Background

An LRE for relation r approximates the model's computation as

$$o \approx \beta W s + b, \tag{1}$$

where s is the hidden state at the subject's final token, o the output state, and W, b are estimated from a small set of reference subjects as the mean Jacobian and mean bias of the model's own map (Hernandez et al., 2024). *Faithfulness* is the rate at which decoding $\beta W s + b$ over the full vocabulary reproduces the *model's own* top-1 prediction—not the gold object. It is therefore a measure of how well an affine map captures what the model does, and is the quantity we predict.

3 A relation-agnostic Jacobian lens

For an entity e and a set of reference entities $\{i\}$ of the same type, we take the Jacobian–vector product of the model's map at each reference and average:

$$v_e \;=\; \frac{1}{n}\sum_i J_i\,(s_e - s_i). \qquad (2)$$

Each J_i is evaluated under a *neutral* prompt, `"{} is known for"`, identical for every relation. We decode v_e over the full vocabulary and record the rank of m_e, the model's own top-1 answer for relation r. Writing $\mathrm{rk}_{v_e}(m_e)$ for that rank, the diagnostic for a relation is

$$A_r \;=\; -\,\mathrm{median}_e\,\log_{10}\mathrm{rk}_{v_e}(m_e), \qquad (3)$$

so higher A_r means the lens agrees with the model more.

Three properties make A_r non-trivial as a predictor. The prompt is neutral and shared, so no relation template enters. The differences in Eq. 2 are subject-minus-subject—*France* − *Turkey*, never *France* − *Paris*—so no (subject, object) pair is used in estimation. And the layer is fixed at 17 for all relations; tuning it per relation would itself be relation-specific information.

Relation-specificity enters in exactly one place: the *target*. The model's top-1 is taken under the same 8-shot in-context prompt used to measure faithfulness, so the two quantities are commensurable.

4 Experimental setup

Model. Olmo-3-7B, `float32` with eager attention (the double-backward used for the JVP is not supported by fused attention kernels).

Relations. The LRE dataset (Hernandez et al., 2024) contains 47 relations. Our prompt presupposes an entity subject, so we fixed a subject-type filter *in advance* using the dataset's declared `domain_name` field: the 28 relations with named-entity subjects (*person*, *country*, *name*, *company*, *landmark*, *superhero*, *city*, *pokemon*, *product*, *star*). Excluded are relations over words as words (*verb*, *adjective*, *word*) and over abstracts (*occupation*, *task*, *characteristic*, *degree*), for which `"give is known for"` is a category error. No relation was dropped after seeing a result.

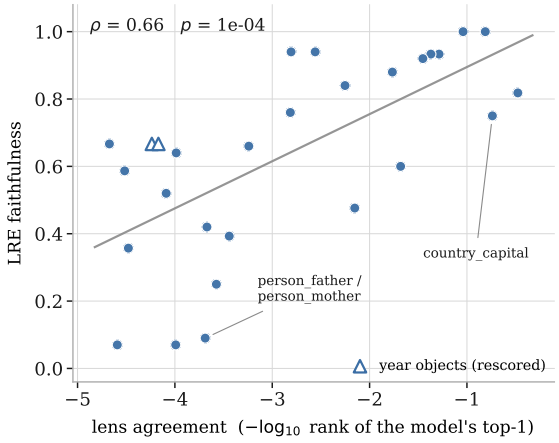


Figure 1: LRE faithfulness against the relation-agnostic lens score, 28 relations. The lens sees no relation template, no objects, and no (subject, object) pairs, and uses one fixed layer throughout. Triangles mark the two relations with numeric objects, rescored as described in §7.

Measurement. Faithfulness is recomputed for all 28 relations under a single protocol (8 in-context examples, 8 reference subjects, a 6×6 grid over layer and β selected on the mean). The lens uses 8 reference subjects and up to 20 scored subjects per relation (11–20, depending on how many the relation has once in-context and reference subjects are removed).

5 Results

Figure 1 shows the result. The relation-agnostic score predicts LRE faithfulness at Spearman $\rho = 0.658$ ($p = 1.4 \times 10^{-4}$; permutation null 0.0003 over 10^4 shuffles).

The correlation is not carried by cardinality. Relations differ greatly in how many distinct objects they have; k, the mean number of subjects per object, ranges from 1.0 to 189. High k makes faithfulness easier, and indeed k predicts faithfulness on its own ($\rho = 0.43$, $p = 0.02$). But k is unrelated to the lens score ($\rho = -0.03$, $p = 0.87$), and partialling it out *raises* the correlation to $\rho = 0.744$ ($p = 10^{-5}$).

Nor by answer frequency. A relation whose answers are frequent tokens might be easy for both measures. We estimate the model's prior over each answer token by averaging its log-probability across ten neutral contexts. This predicts neither faithfulness ($\rho = -0.03$) nor the lens score ($\rho = -0.02$), and partialling it out leaves the cor-

relation unchanged at 0.658; with both covariates, 0.746.

Nor by one domain. Eleven of the 28 relations have *person* subjects, raising the worry that one domain drives the result. Holding out each subject domain in turn leaves $\rho \in [0.570, 0.700]$, median 0.656.

A within-domain check. PERSON_FATHER and PERSON_MOTHER are separate relations over near-identical subject sets. They receive almost identical faithfulness (0.09, 0.07) and almost identical lens scores (-3.69, -3.99), without the lens being told which relation it is looking at.

6 Where the diagnostic fails

Two relations sit well above the trend: NAME_GENDER (faithfulness 0.67, lens -4.67) and PERSON_UNIVERSITY (0.59, -4.52). Both are cases where the model can answer but a neutral "known for" readout has no reason to surface the answer: a first name does not evoke a gender word, and few people are chiefly known for their university. This is the diagnostic's main systematic weakness—it measures what an entity is *salient* for, which is not always what a relation asks for.

7 A first-token scoring artifact

Both relations with numeric objects (PRESIDENTS_BIRTH_YEAR, PRESIDENTS_ELECTION_YEAR) initially scored faithfulness 1.000. This was spurious. In this tokenizer `" 1743"` segments as `[' ', '174', '3']`: digits never merge with a preceding space. First-token scoring therefore compared the model's top-1 against a bare space token on both sides, and faithfulness 1.000 meant only that the LRE and the model both emit a space.

The fix is to place the space in the prompt, so that the query ends with it and the next token is content. Rescored, the two relations give faithfulness 0.67 and lens scores of -4.24 and -4.17; the model does predict the year correctly once the space is consumed. Including the uncorrected values would have depressed the reported correlation from 0.658 to 0.427.

We flag this because the failure is silent: it produces a perfect score rather than an obvious error, and it will affect any first-token evaluation on numeric or unit-prefixed answers.

8 Limitations

The study covers one model and 28 relations. The relationship is correlational: we show that a relation-agnostic readout predicts LRE faithfulness, not that the representation it reads is what the LRE exploits: an intervention experiment is the natural next step. Seven of the 28 relations have fewer than ten test items after the in-context and reference subjects are removed, so their faithfulness estimates are coarse (for three of them, quantized to thirds); a rotating-split protocol would sharpen these. Finally, the diagnostic is defined only for relations whose subjects are entities, which excludes 19 of the dataset's 47 relations.

9 Conclusion

Whether a relation is linearly decodable from a language model is substantially predictable without the relation. A Jacobian lens estimated from subject differences under a neutral prompt, at a fixed layer, ranks the model's own answer highly for exactly those relations whose LREs turn out to be faithful. This locates linear decodability in what a subject's representation already carries, and suggests that the wide spread of LRE faithfulness across relations is less about the relations than about the entities they are asked about.

References

Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. 2024. Linearity of relation decoding in transformer language models. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

A All relations

Table 1 gives faithfulness and the lens score for every relation in the study.

relation	k	faith.	A_r	relation	k	faith.	A_r
country_language*	1.7	1.00	-1.04	presidents_election_year†	1.1	0.67	-4.17
country_largest_city*	1.0	1.00	-0.81	product_by_company	17.4	0.66	-3.24
landmark_on_continent	189.4	0.94	-2.81	person_plays_position_in_sport	68.0	0.64	-3.99
person_native_language	30.6	0.94	-2.56	person_band_lead_singer*	1.0	0.60	-1.68
name_birthplace	3.9	0.93	-1.29	person_university	1.3	0.59	-4.52
name_religion	6.2	0.93	-1.37	company_hq	4.1	0.52	-4.09
person_plays_pro_sport	63.6	0.92	-1.45	superhero_person	1.0	0.48	-2.15
person_plays_instrument	85.5	0.88	-1.77	star_constellation	11.7	0.42	-3.67
landmark_in_country	9.2	0.84	-2.25	pokemon_evolutions	1.0	0.39	-3.44
city_in_country	1.3	0.82	-0.48	country_currency	1.3	0.36	-4.48
person_occupation	26.5	0.76	-2.81	superhero_archnemesis	1.1	0.25	-3.57
country_capital_city*	1.0	0.75	-0.74	person_father	1.0	0.09	-3.69
name_gender*	9.5	0.67	-4.67	company_ceo	1.0	0.07	-4.59
presidents_birth_year†	1.3	0.67	-4.24	person_mother	1.0	0.07	-3.99

Table 1: All 28 relations, sorted by LRE faithfulness. k is the mean number of subjects per distinct object; A_r is the relation-agnostic lens score, higher meaning closer agreement with the model. †rescored for the numeric-object artifact (§7); *fewer than ten test items, so faithfulness is coarsely quantized.